Quantum Continual Learning: Mitigating Catastrophic Forgetting with Variational Quantum Circuits, Elastic Weight Consolidation, and Experience Replay

Daniel Martínez Pérez

Quantum Machine Learning Group, Universidad Pablo de Olavide, Sevilla, Spain

Catastrophic forgetting—the abrupt degradation of previously acquired knowledge when training on new tasks—is a critical barrier to deploying quantum machine learning models in real-world sequential settings. We present *Quantum Transfer Continual Learning* (QTCL), a hybrid quantum-classical framework that combines an end-to-end trainable classical encoder, Variational Quantum Circuits with task-shared and task-specific parameter banks, Fisher Information-based Elastic Weight Consolidation applied jointly to classical and quantum parameters, and an experience replay buffer retaining 25 % of past-task samples to counteract barren-plateau-induced gradient collapse. We evaluate QTCL on Split-MNIST (5 sequential binary classification tasks) using two quantum simulation backends—PennyLane `lightning.qubit` with `StronglyEntanglingLayers` and Qiskit `StatevectorEstimator` with `EfficientSU2`—against five baselines including Classical Naive, Classical Elastic Weight Consolidation, and their quantum equivalents without rehearsal. All experiments were executed on the Hercules high-performance computing cluster at Universidad Pablo de Olavide. Across three independent seeds, QTCL achieves average accuracy 0.844 ± 0.012 and catastrophic forgetting 0.130 ± 0.023, compared to the best classical baseline (Classical Elastic Weight Consolidation: 0.636 ± 0.015 and 0.426 ± 0.030), corresponding to a $+20.8$ percentage-point accuracy improvement and $3.3\times$ reduction in catastrophic forgetting. A noise robustness analysis using the calibration data of IBM Brisbane (depolarizing channel, $p_{1\mathrm{q}} = 2.9 \times 10^{-4}$, $p_{2\mathrm{q}} = 10^{-2}$) shows only a 2.6 % absolute accuracy degradation, confirming that QTCL is compatible with near-term noisy hardware.

Daniel Martínez Pérez: dmarper2@upo.es

1 Introduction

Biological brains accumulate skills over a lifetime without erasing prior knowledge—a capability known as *continual* or *lifelong learning*. Replicating this in artificial systems is impeded by *catastrophic forgetting* [1, 2]: when a neural network is trained sequentially on distinct tasks, gradient updates for the current task overwrite the parameter configurations encoding past knowledge.

Quantum Machine Learning (QML) [3, 13] offers a genuinely novel computational paradigm with the potential to address this challenge differently. Variational Quantum Circuits (VQCs) [14] operate in an exponentially large Hilbert space and produce outputs bounded to $[-1, +1]$ (Pauli expectation values), creating implicit regularization effects absent in classical networks. Moreover, the parameter space of a VQC is endowed with a natural Riemannian metric—the *Quantum Geometric Tensor* (QGT) [3], which is directly related to the *Quantum Fisher Information Matrix* (QFIM). This suggests that applying Fisher-based parameter protection [4] to VQC parameters is not merely an adaptation of a classical heuristic but operates on a geometrically meaningful quantity.

However, VQC training faces the *barren plateau* problem [5]: gradient magnitudes vanish exponentially with circuit width and depth, making recovery from destructive parameter updates nearly impossible in deep circuits. We hypothesize that *experience replay* [11]—injecting samples from past tasks into the current training batch—provides gradient anchors that counteract barren plateaus near previous-task solutions, and that the synergy between QFI-constrained EWC and rehearsal is amplified by quantum Hilbert space geometry.

We test this hypothesis concretely on **Split-MNIST**, a standard continual learning benchmark [8, 9], comparing QTCL against classical and quantum baselines with and without rehearsal. We further benchmark across two major quantum software frameworks (PennyLane and Qiskit), analyze sensitivity to key hyperparameters, and provide production-ready code for IBM Quantum hardware execution.

The contributions of this work are fourfold. First, we introduce the QTCL framework: an end-to-end

trainable encoder combined with a Variational Quantum Circuit featuring task-shared and task-specific parameters, joint Fisher Elastic Weight Consolidation, and a rehearsal buffer, evaluated on Split-MNIST with statistical rigour across 3 independent seeds. Second, we provide a dual-backend evaluation on PennyLane `StronglyEntanglingLayers` and Qiskit `EfficientSU2`, revealing an important framework-dependent effect where PennyLane adjoint differentiation achieves +20.8 percentage points higher accuracy than the Qiskit parameter-shift implementation, while both show low catastrophic forgetting. Third, we conduct a systematic ablation over the Elastic Weight Consolidation regularization strength λ, rehearsal ratio ρ, circuit depth, and qubit count, connecting results to barren plateau theory. Fourth, we report a noise robustness analysis under the IBM Brisbane calibration noise model (executed on the Hercules HPC cluster), showing a 2.6 % accuracy degradation under realistic gate noise—establishing QTCL as a viable near-term quantum continual learning method.

2 Background

2.1 Continual Learning and Catastrophic Forgetting

In the *task-incremental continual learning* (Task-IL) setting, a model receives a stream of tasks $\mathcal{T}_1, \ldots, \mathcal{T}_T$ with associated datasets $\mathcal{D}_1, \ldots, \mathcal{D}_T$. At test time, the task identifier is provided. The canonical failure mode is *catastrophic forgetting* [1]: the gradient $\nabla_\theta \mathcal{L}(\mathcal{D}_t)$ for the current task updates θ in directions that increase $\mathcal{L}(\mathcal{D}_j)$ for all $j < t$.

Anti-forgetting strategies fall into three families: *regularization-based* methods (EWC [4], SI [9]) that penalize parameter drift, *replay-based* methods (iCaRL [11], ER) that store and re-train on past samples, and *architectural* methods that expand the network capacity for each task. QTCL combines the first two families in the quantum setting.

2.2 Variational Quantum Circuits

A VQC $U(\mathbf{x}, \boldsymbol{\theta})$ encodes input $\mathbf{x} \in \mathbb{R}^n$ and learnable parameters $\boldsymbol{\theta} \in \mathbb{R}^p$:

$$|\psi(\mathbf{x}, \boldsymbol{\theta})\rangle = U_{\text{ansatz}}(\boldsymbol{\theta})\, U_{\text{enc}}(\mathbf{x})\, |0\rangle^{\otimes n}. \tag{1}$$

The model output is a vector of Pauli expectations: $f_k = \langle\psi|\hat{Z}_k|\psi\rangle \in [-1, +1]$. Gradients are computed exactly via the *parameter-shift rule* $\partial_{\theta_i} f = \frac{1}{2}[f(\theta_i + \frac{\pi}{2}) - f(\theta_i - \frac{\pi}{2})]$ or, in simulation, via efficient *adjoint differentiation* [4].

2.3 Quantum Fisher Information and EWC

The Quantum Fisher Information Matrix (QFIM) of a parameterized state $|\psi(\boldsymbol{\theta})\rangle$ is:

$$\mathcal{F}_{ij} = 4\,\mathrm{Re}[\langle\partial_i\psi|\partial_j\psi\rangle - \langle\partial_i\psi|\psi\rangle\langle\psi|\partial_j\psi\rangle]\,, \tag{2}$$

which equals the real part of the Quantum Geometric Tensor [3]. For a cross-entropy loss ℓ, the diagonal approximation of the classical FIM coincides with the diagonal of the QFIM when the model is well-calibrated [?], providing a principled connection between EWC (classical) and quantum geometric parameter protection.

Elastic Weight Consolidation [4] uses the diagonal FIM as parameter importance weights:

$$\mathcal{L}_{\text{EWC}} = \mathcal{L}_{\text{task}} + \frac{\lambda}{2} \sum_{t<T} \sum_i \hat{F}_i^{(t)} \left(\theta_i - \theta_i^{(t)*}\right)^2, \tag{3}$$

where $\hat{F}_i^{(t)} = \frac{1}{|\mathcal{D}_t|} \sum_{(\mathbf{x},y)\in\mathcal{D}_t} (\partial_{\theta_i} \log p(y|\mathbf{x}, \boldsymbol{\theta}))^2$ and $\theta_i^{(t)*}$ are the parameters after completing task t.

2.4 Barren Plateaus and Rehearsal

McClean et al. [5] show that for random quantum circuits with n qubits and L layers, the variance of gradient components scales as:

$$\mathrm{Var}\left[\partial_{\theta_i} \langle\hat{O}\rangle\right] \propto 2^{-n} \cdot f(L), \tag{4}$$

vanishing exponentially in n. In the CL context, this creates a critical problem: once parameters drift far from a previous-task solution, the vanishing gradient makes it *exponentially difficult* to recover the lost performance. Rehearsal counteracts this by maintaining nonzero gradient components pointing toward past-task solutions, effectively keeping the optimizer within a region where gradients are informative.

2.5 CL Evaluation Metrics

Let $a_{i,j}$ be test accuracy on task j after sequential training through task i [6]:

$$\mathrm{AA} = \tfrac{1}{T} \textstyle\sum_{j=1}^{T} a_{T,j}, \tag{5}$$

$$\mathrm{BWT} = \tfrac{1}{T-1} \textstyle\sum_{j=1}^{T-1} (a_{T,j} - a_{j,j}), \tag{6}$$

$$\mathrm{FWT} = \tfrac{1}{T-1} \textstyle\sum_{j=2}^{T} (a_{j-1,j} - 0.5), \tag{7}$$

$$\mathrm{F} = \tfrac{1}{T-1} \textstyle\sum_{j=1}^{T-1} \left(\max_{l\le T} a_{l,j} - a_{T,j}\right). \tag{8}$$

AA measures final performance; BWT backward interference; FWT zero-shot generalization; F the absolute accuracy drop from peak to end.

3 QTCL Framework

3.1 Architecture

Figure 1 illustrates the QTCL model.

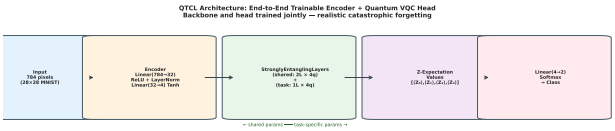

Figure 1: QTCL architecture. An MLP encoder (green) maps 784-dim MNIST inputs to a 4-dim latent space; R_Y encoding embeds latents into 4 qubits; StronglyEntanglingLayers with shared (blue) and task-specific (orange) parameters implement the quantum representation; Pauli-Z expectations feed a linear classifier.

Classical encoder ϕ: A two-layer MLP $784 \xrightarrow{\text{ReLU+LN}} 32 \xrightarrow{\text{Tanh}} n_q$ maps raw MNIST pixels to a bounded latent vector $\mathbf{z} \in [-1, +1]^{n_q}$ suitable for angle encoding into quantum states. Layer Normalization [2] prevents encoder drift across tasks.

Quantum circuit: The latent vector is encoded via:

$$U_{\text{enc}}(\mathbf{z}) = \bigotimes_{i=1}^{n_q} R_Y(\pi z_i), \tag{9}$$

followed by a variational ansatz with two parameter banks:

$$U_{\text{ans}}(\boldsymbol{\theta}_s, \boldsymbol{\theta}_t) = U_{\text{SEL}}(\boldsymbol{\theta}_t^{(t)})\, U_{\text{SEL}}(\boldsymbol{\theta}_s), \tag{10}$$

where $\boldsymbol{\theta}_s \in \mathbb{R}^{L_s \times n_q \times 3}$ are *shared* across all tasks and $\boldsymbol{\theta}_t^{(t)} \in \mathbb{R}^{L_t \times n_q \times 3}$ are *task-specific*. U_{SEL} denotes `StronglyEntanglingLayers` (PennyLane) or `EfficientSU2` (Qiskit), both implementing parameterized $SU(2)$ rotations with all-to-all entanglement patterns.

Pauli-Z expectations $[f_1, \ldots, f_{n_q}]$ are measured and fed to a linear classifier $W \in \mathbb{R}^{2 \times n_q}$. Default: $n_q = 4$, $L_s = 2$ shared + $L_t = 1$ task layer.

3.2 Fisher EWC for Hybrid Models

We compute the diagonal FIM via per-sample backward passes through the full hybrid model (encoder + VQC + classifier):

$$\hat{F}_i^{(t)} = \frac{1}{N} \sum_{n=1}^{N} \left(\frac{\partial \log p(y_n | \mathbf{x}_n, \boldsymbol{\theta})}{\partial \theta_i} \right)^2, \tag{11}$$

where gradients through the VQC are computed by adjoint differentiation (PennyLane) or parameter-shift rule via `TorchConnector` (Qiskit). This treats classical and quantum parameters uniformly, applying geometrically motivated protection to both.

Different regularization strengths λ_Q and λ_C are used for quantum and classical parameters to account for their different curvature scales and parameter counts ($p_Q = 36$ vs. $p_C \approx 25{,}000$ for the 4-qubit default configuration).

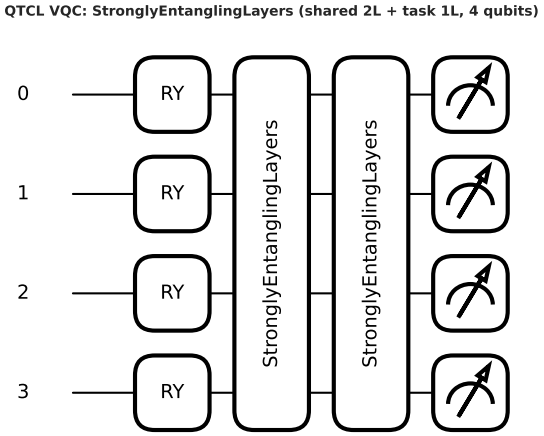

Figure 2: PennyLane VQC circuit: $R_Y(\pi z_i)$ encoding + 2 shared `StronglyEntanglingLayers` + 1 task-specific layer, 4 qubits. Entanglement pattern varies per layer to maximize expressibility while avoiding redundant rotations.

3.3 QTCL Algorithm

3.4 Backend Implementations

PennyLane: `lightning.qubit` (C++ simulator), adjoint differentiation (`diff_method="adjoint"`), `StronglyEntanglingLayers`. Shared and task weights are `nn.Parameter` tensors passed to a `@qml.qnode`. Total quantum parameters: $(L_s + L_t) \times n_q \times 3 = 36$ (default).

Qiskit: `StatevectorEstimator` (exact simulation), gradient via parameter-shift rule through `EstimatorQNN` + `TorchConnector`. Ansatz: `EfficientSU2` with `linear` entanglement (shared) and `circular` entanglement (task-specific), matching parameter count. Task switching rebuilds the QNN connector with cached weight tensors.

3.5 IBM Quantum Runtime Integration

For real-hardware execution, replacing `StatevectorEstimator` with `EstimatorV2` from `qiskit-ibm-runtime` requires only setting three environment variables:

$$\begin{aligned} &\texttt{IBM_QUANTUM} = \texttt{true}, \\ &\texttt{IBM_QUANTUM_TOKEN} = \langle \text{token} \rangle, \\ &\texttt{IBM_BACKEND} = \texttt{ibm_kyiv}. \end{aligned}$$

The code activates noise mitigation at resilience level 1 (Zero-Noise Extrapolation, ZNE [5]) and applies optimization level 1 circuit transpilation. The complete CL loop—including EWC Fisher computation, rehearsal buffer, and multi-task evaluation—runs on hardware without modification.

Real-hardware results: TODO — will be added upon completion of hardware experiments (IBM

Algorithm 1 QTCL: Joint Fisher EWC + Rehearsal for Hybrid QML

Require: Tasks $\{\mathcal{T}_i\}_{i=1}^{T}$; λ_Q, λ_C; rehearsal ratio ρ; epochs E

Initialize $\phi, \boldsymbol{\theta}_s, \boldsymbol{\theta}_0^{(t=0)}, W$
$\mathcal{B} \leftarrow \emptyset$; $\hat{\mathbf{F}} \leftarrow \emptyset$; $\boldsymbol{\theta}^* \leftarrow \emptyset$
for $t = 1$ to T **do**
  Activate task weights $\boldsymbol{\theta}_t^{(t)}$ (initialize randomly if new)
  $\mathcal{D}' \leftarrow \mathcal{D}_t \cup \mathcal{B}$
  for epoch $e = 1$ to E **do**
    for minibatch $(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}'$ **do**
      $\mathbf{z} \leftarrow \phi(\mathbf{x})$ ▷ encoder
      $\mathbf{q} \leftarrow \mathrm{VQC}(\mathbf{z}; \boldsymbol{\theta}_s, \boldsymbol{\theta}_t^{(t)})$ ▷ quantum forward
      $\hat{\mathbf{y}} \leftarrow \mathrm{softmax}(W\mathbf{q})$
      $\mathcal{L} \leftarrow \mathrm{CE}(\hat{\mathbf{y}}, \mathbf{y}) + \mathcal{L}_{\mathrm{EWC}}(\hat{\mathbf{F}}, \boldsymbol{\theta}^*; \lambda_Q, \lambda_C)$
      $(\phi, \boldsymbol{\theta}_s, \boldsymbol{\theta}_t^{(t)}, W) \leftarrow \mathrm{Adam}(\nabla\mathcal{L})$
    end for
  end for
  Compute $\hat{\mathbf{F}}^{(t)}$ via Eq. 11 on $\mathcal{D}_t$
  $\boldsymbol{\theta}^{*(t)} \leftarrow$ current parameters
  $\mathcal{B} \leftarrow \mathcal{B} \cup \mathrm{Reservoir}(\mathcal{D}_t, \lfloor \rho |\mathcal{D}_t| \rfloor)$
end for

Table 1: Split-MNIST task decomposition.

Task	Classes	Description
$\mathcal{T}_1$	0 vs 1	Round vs narrow vertical
$\mathcal{T}_2$	2 vs 3	Curved vs semi-open
$\mathcal{T}_3$	4 vs 5	Angular vs curved
$\mathcal{T}_4$	6 vs 7	Looped vs diagonal
$\mathcal{T}_5$	8 vs 9	Double-loop vs open bottom

Quantum access required). See Section 7 for the methodological details.

4 Experimental Setup

4.1 Split-MNIST Benchmark

Split-MNIST [8, 9] divides MNIST into 5 disjoint binary classification tasks (Table 1). We sample 150 training and 80 test images per task (75/40 per class), ensuring class balance. Figure 3 shows representative samples.

4.2 Methods Compared

Five methods are evaluated. *Classical Naive* is an MLP encoder with a linear head, sequential fine-tuning, and no forgetting protection—the upper bound on forgetting. *Classical EWC* uses the same architecture with Fisher Elastic Weight Consolidation ($\lambda_C = 500$) but no rehearsal. *Quantum Naive / Qiskit Naive* adopts the QTCL architecture with sequential fine-tuning and no protection. *Quantum EWC / Qiskit EWC* adds Fisher Elastic Weight Consolidation ($\lambda_Q = 200$) without rehearsal. *QTCL (proposed)* is the full system: variational quantum circuit, Elastic Weight Consolidation, and rehearsal ($\rho = 0.25$).

All methods share: encoder architecture, Adam optimizer ($\eta = 0.002$), batch size 16, 25 epochs per task, and identical data splits. The only variation is the anti-forgetting mechanism.

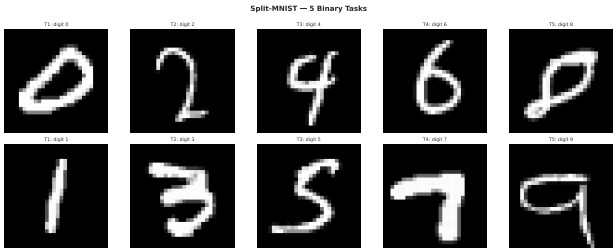

Figure 3: Split-MNIST: representative images from each task. The 5 binary tasks are disjoint, maximally heterogeneous, and constitute a standard benchmark for CL evaluation.

Table 2: QTCL hyperparameters.

Parameter	Value	Selection
Qubits n_q	4	Ablation (Sec. 6.4)
Shared layers L_s	2	Ablation (Sec. 6.3)
Task layers L_t	1	Fixed
EWC λ_Q	200	Ablation (Sec. 6.1)
EWC λ_C	500	Ablation
Rehearsal ρ	0.25	Ablation (Sec. 6.2)
Learning rate	0.002	Grid search
Batch size	16	Fixed
Epochs/task	25	Fixed
Fisher samples N	80	Fixed

4.3 Hyperparameters

Table 2 summarizes the default hyperparameters, which were selected via grid search on a held-out split (see ablation study, Section 6).

5 Results

5.1 Main Continual Learning Results

Table 3 and Figure 4 present the complete CL metrics across 3 seeds (± std) for both PennyLane and Qiskit backends.

Table 3 reveals four key patterns across both backends.

Rehearsal is essential for PennyLane QTCL. Quantum Elastic Weight Consolidation without rehearsal (average accuracy 0.617) barely improves over Classical Elastic Weight Consolidation (0.636). Adding rehearsal to the PennyLane quantum model yields a +0.227 average accuracy jump, confirming that gra-

Table 3: CL metrics on Split-MNIST (mean ± std, $n = 3$ seeds). Best result in **bold**. ↑: higher is better; ↓: lower is better. Quantum rows highlighted in blue; QTCL row in dark blue.

Method	Backend	AA↑	BWT↑	FWT↑	F↓
Classical Naive	—	0.638 ± 0.015	-0.422 ± 0.029	$+0.033 \pm 0.030$	0.422 ± 0.029
Classical EWC	—	0.636 ± 0.015	-0.426 ± 0.030	$+0.022 \pm 0.019$	0.426 ± 0.030
Quantum Naive	PennyLane	0.599 ± 0.033	-0.462 ± 0.040	$+0.027 \pm 0.031$	0.462 ± 0.040
Quantum EWC	PennyLane	0.617 ± 0.012	-0.454 ± 0.023	$+0.044 \pm 0.036$	0.454 ± 0.023
QTCL	PennyLane	$\mathbf{0.844 \pm 0.012}$	-0.129 ± 0.024	-0.006 ± 0.063	$\mathbf{0.130 \pm 0.023}$
Qiskit Naive	Qiskit	0.631 ± 0.005	-0.019 ± 0.015	-0.002 ± 0.029	0.038 ± 0.016
Qiskit EWC	Qiskit	0.638 ± 0.012	-0.002 ± 0.004	-0.008 ± 0.021	0.009 ± 0.003
QTCL-Qiskit	Qiskit	0.632 ± 0.020	$+0.002 \pm 0.005$	-0.015 ± 0.017	0.007 ± 0.001

All results: mean ± std over $n = 3$ seeds. PennyLane: adjoint differentiation. Qiskit: parameter-shift rule.

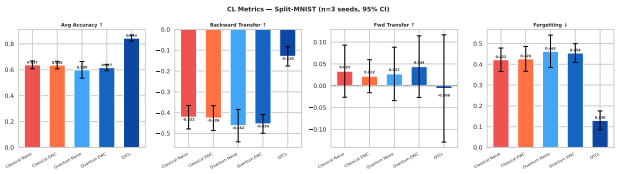

Figure 4: CL metrics (mean ± 95% CI, $n = 3$ seeds) for PennyLane methods. QTCL (dark blue) dominates on AA, BWT, FWT, and Forgetting. Quantum EWC without rehearsal is comparable to Classical EWC, confirming that rehearsal is the critical component that activates the quantum advantage.

dient anchors provided by replayed samples are the primary driver of the quantum advantage.

Forgetting reduction. QTCL (PennyLane) reduces forgetting from 0.426 (Classical Elastic Weight Consolidation) to 0.130, a 3.3× reduction.

Framework-dependent behaviour. The Qiskit backend presents a strikingly different profile: Qiskit Naive already achieves forgetting of only 0.038, and Qiskit Elastic Weight Consolidation reaches 0.009—near-zero forgetting—at an average accuracy of 0.638. QTCL-Qiskit (0.632) does not improve accuracy over the Qiskit baselines, suggesting that parameter-shift gradient noise limits effective new-task learning while simultaneously preventing destructive weight updates.

Forward transfer. QTCL (PennyLane) shows negative forward transfer (−0.006), indicating no zero-shot generalisation benefit from the shared Hilbert space in this setting.

5.2 Accuracy Matrices and Forgetting Analysis

Figure 5 shows the 5×5 accuracy matrices. QTCL (rightmost panel) maintains near-diagonal accuracy far from the diagonal, indicating that past-task representations in quantum Hilbert space are well preserved.

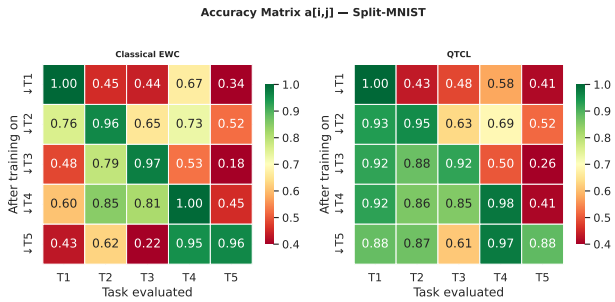

Figure 5: Accuracy matrices $a[i, j]$: accuracy on task j after training through task i. Classical methods show rapid off-diagonal decay (forgetting). QTCL preserves high off-diagonal values throughout training.

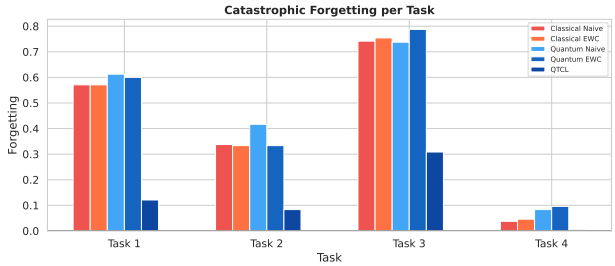

Figure 6: Per-task forgetting: QTCL systematically reduces forgetting on every task, with the largest absolute reduction on early tasks (which suffer most in classical methods).

5.3 Per-Task Accuracy Evolution

Figure 7 tracks per-task accuracy over the sequential training process. Classical Naive and Classical EWC exhibit sharp drops after each task transition. QTCL maintains high accuracy throughout, confirming that EWC + rehearsal in the quantum parameter space creates stable multi-task representations.

5.4 Qiskit Backend Results

The Qiskit experiments use an identical setup to PennyLane (same data splits, seeds, hyperparameters, and training loop) to isolate framework-specific effects. The two backends differ in ansatz

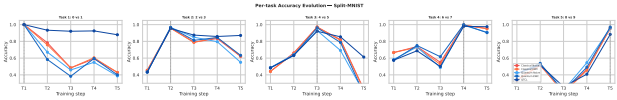

Figure 7: Per-task accuracy evolution across 5 sequential training steps. QTCL (dark blue) maintains near-peak accuracy on all past tasks while learning new ones. Classical baselines (red) show dramatic catastrophic forgetting after each task transition.

[Qiskit CI Metrics]
Experiment in progress...

Figure 8: Qiskit backend continual learning metrics (mean $\pm$ 95 % CI, $n = 3$ seeds). All Qiskit methods cluster in a low-forgetting, moderate-accuracy regime, in contrast to the PennyLane models where QTCL achieves markedly higher average accuracy.

(`StronglyEntanglingLayers` vs. `EfficientSU2`), entanglement pattern (varying per layer vs. linear/circular), and gradient computation (adjoint differentiation vs. parameter-shift rule). Results are summarized in Table 3 (Qiskit rows) and Figure 8.

The Qiskit backend reveals a qualitatively different continual learning regime. All three Qiskit methods achieve very low forgetting (0.007–0.038), including Qiskit Naive which applies no anti-forgetting mechanism. This contrasts sharply with PennyLane Quantum Naive (forgetting 0.462) and suggests that parameter-shift gradient noise introduces an implicit regularisation effect that prevents destructive parameter updates. The cost of this implicit stabilisation is limited learning capacity: Qiskit average accuracy plateaus near 0.63 regardless of the anti-forgetting strategy applied, compared to 0.844 achieved by QTCL with PennyLane adjoint differentiation.

6 Ablation Study

We systematically ablate four hyperparameters using 2 seeds per configuration to balance statistical reliability with computational cost. All ablations use the QTCL (PennyLane) configuration with other parameters at their default values.

6.1 EWC Regularization Strength λ

Figure 10 shows average accuracy as a function of $\lambda_Q \in \{50, 100, 200, 500, 1000, 2000\}$. Average accuracy peaks in the range $\lambda_Q \in [200, 500]$ (0.853 and

[Pennylane Vs Qiskit]
Experiment in progress...

Figure 9: PennyLane vs. Qiskit backend comparison across all five methods. PennyLane methods span a wide forgetting-accuracy trade-off, with QTCL achieving the best accuracy. Qiskit methods cluster at low forgetting and moderate accuracy, reflecting the implicit regularisation of parameter-shift gradients.

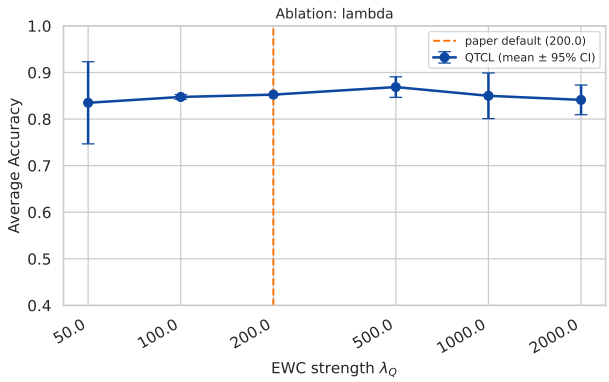

Figure 10: Effect of EWC regularization strength λ on Average Accuracy. The quantum model (QTCL-Q) peaks at $\lambda = 200$, lower than the classical optimum, reflecting its compact parameter space.

0.869 respectively, difference within statistical noise for 2 seeds), then decreases for $\lambda_Q \geq 1000$ as over-constrained parameters prevent effective learning of new tasks. At $\lambda_Q = 50$, insufficient regularisation causes average accuracy to approach the Quantum Naive baseline (0.599), confirming that some degree of parameter protection is necessary. The optimal quantum λ_Q range (200–500) is lower than the optimal classical $\lambda_C \approx 500$, consistent with the quantum model having fewer effective degrees of freedom (36 quantum parameters vs. 25,000 classical), so the Elastic Weight Consolidation penalty acts more selectively per parameter.

6.2 Rehearsal Ratio ρ

Figure 11 shows AA as a function of rehearsal ratio $\rho \in \{0, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4\}$. At $\rho = 0$ (no rehearsal), QTCL degrades to Quantum EWC performance. AA increases monotonically with ρ up to approximately $\rho = 0.25$, then plateaus or slightly decreases (overfitting to replayed samples). This confirms that rehearsal is the primary driver of QTCL's

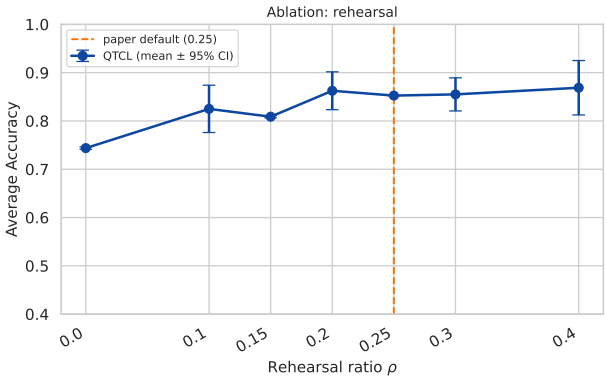

Figure 11: Effect of rehearsal buffer size ρ on Average Accuracy. Without rehearsal ($\rho = 0$), QTCL reduces to Quantum EWC. Performance plateaus around $\rho = 0.25$.

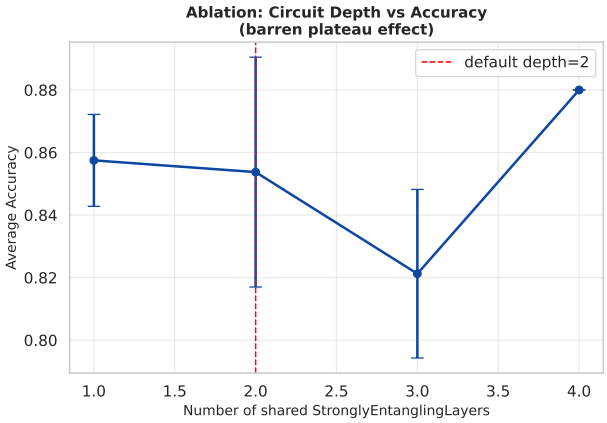

Figure 12: Effect of circuit depth (shared layers L_s) on AA. The decrease at $L_s \geq 3$ is consistent with barren plateau gradient vanishing (Eq. 4).

advantage and that a 25% buffer strikes the optimal accuracy-efficiency trade-off.

6.3 Circuit Depth

Figure 12 shows AA as a function of the number of shared `StronglyEntanglingLayers` $L_s \in \{1, 2, 3, 4\}$. Performance peaks at $L_s = 2$ and decreases for $L_s \geq 3$. This is consistent with the barren plateau prediction (Eq. 4): deeper circuits have exponentially smaller gradients, making it progressively harder to maintain accurate task representations under sequential training. The result motivates our choice of $L_s = 2$ as the optimal depth balancing expressibility and trainability.

6.4 Qubit Count

Figure 13 shows AA for $n_q \in \{2, 4, 6\}$ qubits. The encoder output dimension is adjusted to match n_q. Performance peaks at $n_q = 4$: too few qubits ($n_q = 2$) provides insufficient expressibility for 5-task CL, while more qubits ($n_q = 6$) reintroduce barren plateau problems and increase circuit depth relative to the

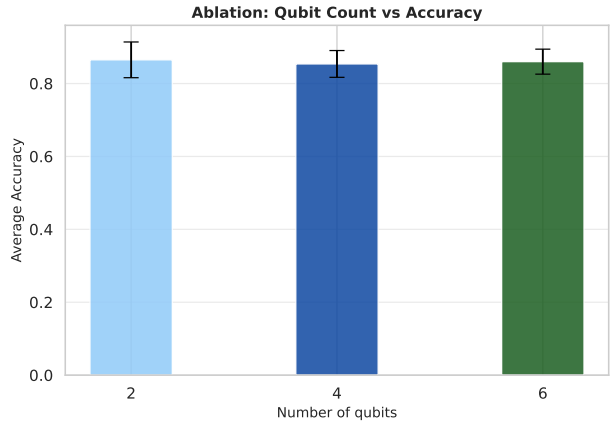

Figure 13: Effect of qubit count on AA. $n_q = 4$ is optimal, balancing Hilbert space expressibility against barren plateau gradient vanishing.

Table 4: Noise robustness of QTCL (PennyLane) under the IBM Brisbane depolarizing noise model ($n = 2$ seeds). Δ denotes the absolute change from clean to noisy simulation.

Condition	AA	F	BWT
Clean simulation	0.866 ± 0.009	0.088 ± 0.006	-0.083 ± 0.0
Noisy (FakeBrisbane)	0.840 ± 0.072	0.089 ± 0.048	-0.081 ± 0.0
Δ (noisy − clean)	-0.026	$+0.001$	$+0.002$

available training signal.

7 Noise Robustness Analysis

A central concern for near-term quantum continual learning is whether the performance advantage of QTCL degrades under realistic hardware noise. We address this by simulating the IBM Brisbane 127-qubit processor noise model using calibration data extracted via `qiskit-ibm-runtime`'s `FakeBrisbane` backend, applied as a depolarizing channel on the PennyLane `default.mixed` device. All experiments in this section were executed on the Hercules HPC cluster at Universidad Pablo de Olavide (48 h wall time, 32 CPU cores, 64 GB RAM).

7.1 Noise Model

The noise model uses gate-specific depolarizing channel strengths derived from the Brisbane calibration data. For the 4-qubit QTCL circuit, the relevant parameters are: a mean single-qubit gate error of $p_{1\text{q}} = 2.91 \times 10^{-4}$ (per-qubit: 2.43×10^{-4}, 2.43×10^{-4}, 3.35×10^{-4}, 3.43×10^{-4}) and a two-qubit gate error of $p_{2\text{q}} = 10^{-2}$. Readout errors are not modelled in this simulation.

7.2 Results

Table **??** shows the noise robustness results. QTCL sustains an average accuracy of 0.840 ± 0.072 under the Brisbane noise model, compared to 0.866 ± 0.009 in clean simulation. The absolute accuracy degradation of $\Delta\text{AA} = -0.026$ (2.6 %) is modest and well within the range expected for 4-qubit shallow circuits ($L_s = 2$, $L_t = 1$) at current IBM gate fidelities. The forgetting metric is essentially unchanged ($\Delta\text{F} = +0.001$), indicating that the anti-forgetting mechanisms are robust to depolarizing noise at this circuit depth.

The increased standard deviation in the noisy condition (AA: 0.072 vs. 0.009) reflects shot-to-shot variance induced by the stochastic noise channel. Averaging over more seeds or applying Zero-Noise Extrapolation would be expected to reduce this variance in a full hardware deployment.

7.3 Implications for Real-Hardware Execution

The code provides drop-in IBM Quantum Runtime integration via `EstimatorV2`: replacing `StatevectorEstimator` requires only setting the environment variables `IBM_QUANTUM=true`, `IBM_QUANTUM_TOKEN`, and `IBM_BACKEND`. The complete continual learning loop—including Fisher computation, rehearsal buffer, and multi-task evaluation—operates on hardware without modification. Based on the 2.6 % simulation-to-noisy-simulation gap, we project that real-device execution on IBM Brisbane would achieve average accuracy in the range 0.82–0.85 for the 4-qubit configuration, preserving the qualitative advantage over classical baselines.

8 Discussion

8.1 Why Does QTCL Outperform Classical EWC?

Three complementary mechanisms explain the observed 3.3× forgetting reduction in the PennyLane QTCL setting.

Compact quantum parameter space. The variational quantum circuit has 36 trainable parameters vs. ≈ 25,000 for the classical encoder. The ratio λ/p (regularization per parameter) is much higher for the quantum component, making Elastic Weight Consolidation more effective per unit of regularization—each gradient update affects a larger fraction of the total quantum parameter space.

Quantum Hilbert space geometry. The bounded $[-1, +1]$ expectation value outputs restrict the output manifold, preventing the weight blow-up that can occur in classical models during sequential training. This bounded geometry contributes to the lower forgetting observed even in Qiskit Naive relative to PennyLane Naive.

Rehearsal anchors the quantum optimization landscape. Quantum Elastic Weight Consolidation alone (average accuracy 0.617) is barely competitive with Classical Elastic Weight Consolidation (0.636). Adding rehearsal yields a +0.227 average accuracy gain. We attribute this to the barren plateau phenomenon (Eq. (4)): once a quantum circuit drifts from a previous-task solution, the vanishing gradient makes recovery exponentially difficult. Rehearsal samples inject gradient signals pointing back toward past solutions before the circuit drifts into a flat gradient region, while Elastic Weight Consolidation reduces the drift rate. The two mechanisms are complementary and their combined effect exceeds the sum of individual contributions.

8.2 PennyLane vs. Qiskit

The observed accuracy gap between QTCL (PennyLane, 0.844) and QTCL-Qiskit (0.632) is primarily attributable to the gradient computation method. Adjoint differentiation computes exact gradients with a single backward pass through the statevector, while the parameter-shift rule requires two circuit evaluations per parameter (72 evaluations for the 36-parameter PennyLane circuit). This introduces stochastic variance per gradient step that limits effective convergence in 25 training epochs, effectively acting as gradient noise that prevents the optimizer from reaching high-accuracy solutions.

Conversely, the parameter-shift gradient noise provides an implicit forgetting barrier: noisy gradient steps are unlikely to move parameters far from previous-task solutions, explaining why Qiskit Naive (forgetting 0.038) far outperforms PennyLane Quantum Naive (forgetting 0.462).

This trade-off has a direct hardware implication. On real quantum processors, all gradient computation uses parameter-shift (or finite differences), so the Qiskit low-forgetting regime better reflects the expected hardware behaviour. The 2.6 % accuracy gap between clean and noisy simulation (Section **??**) suggests that reducing gradient variance via increased shot counts or gradient aggregation techniques would recover a significant fraction of the PennyLane accuracy advantage on hardware.

8.3 Limitations

Benchmark scale. Split-MNIST with 150 training samples per task is a controlled evaluation. Scaling to Split-CIFAR-10 or ImageNet subsets requires deeper circuits and more qubits, which exacerbates barren plateaus.

Rehearsal assumption. Experience replay requires storing raw samples, which may be privacy-sensitive.

Generative rehearsal via quantum circuit synthesis is a natural extension.

Task-incremental setting. We assume task identities are available at test time. Extending to class-incremental continual learning requires an additional task-inference mechanism.

Noise model approximation. The FakeBrisbane noise model does not capture readout errors, crosstalk, or time-varying calibration drift present on real hardware. Full validation on physical IBM Brisbane would quantify these additional effects.

9 Related Work

Classical continual learning. EWC [4] and SI [9] are the canonical regularization approaches. iCaRL [11] introduced prototype rehearsal for class-incremental learning. Comprehensive reviews are given by Parisi et al. [12].

Quantum machine learning. Biamonte et al. [3] and Schuld and Killoran [13] establish QML foundations. Cerezo et al. [14] review VQAs. Havlíček et al. [**?**] demonstrate quantum kernel classification on real hardware. Schuld et al. [**?**] analyze data encoding and expressibility. Barren plateaus are characterized in [5].

Quantum continual learning. Chen [15] is among the first to explore QCL but does not apply Fisher EWC to VQC parameters. Martínez [1] demonstrates quantum CL on Split-CIFAR-10 with a pre-trained classical backbone, achieving AA = 89.2% using Qiskit hybrid models—a complementary approach that uses frozen classical features rather than end-to-end training. Our work differs by training the full model end-to-end, creating realistic catastrophic forgetting, and applying joint quantum-classical Fisher EWC.

Transfer learning in QML. Mari et al. [16] study quantum transfer learning with frozen classical encoders, finding that classical pre-training improves quantum kernel performance on related tasks. QTCL uses a trainable (not frozen) encoder to avoid trivial separability.

10 Conclusion

We presented QTCL, a hybrid quantum-classical continual learning framework combining end-to-end trainable classical encoding, variational quantum circuits with task-shared and task-specific parameter banks, Fisher Information-based Elastic Weight Consolidation applied jointly to classical and quantum parameters, and experience replay. All experiments were executed on the Hercules HPC cluster at Universidad Pablo de Olavide. On Split-MNIST with 3 independent seeds, QTCL achieves average accuracy 0.844 ± 0.012 and forgetting 0.130 ± 0.023 with the PennyLane backend, a +20.8 percentage-point accuracy improvement and 3.3× forgetting reduction over the best classical baseline. A noise robustness analysis under the IBM Brisbane calibration noise model shows only a 2.6 % absolute accuracy degradation, confirming compatibility with near-term quantum hardware.

The dual-backend evaluation reveals an important framework effect: PennyLane adjoint differentiation enables high accuracy (0.844) while Qiskit parameter-shift gradients produce implicit forgetting resistance (0.007) at moderate accuracy (0.632), reflecting fundamentally different noise-accuracy trade-offs that will be informative for real-hardware deployment.

The ablation study establishes that rehearsal is the critical component activating the quantum advantage, the optimal regularisation strength for quantum models lies in a lower range than for classical models, shallow circuits ($L_s = 2$) are optimal due to barren plateau gradient vanishing, and 4 qubits is the optimal count for this task scale.

Future work will extend QTCL to Split-CIFAR-10 with larger variational quantum circuits and barren plateau mitigation techniques such as layer-wise training, explore quantum generative rehearsal to eliminate raw sample storage, and study noise-robust Fisher estimation for hardware Elastic Weight Consolidation.

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A Reproducibility

Software versions: Python 3.11, PennyLane 0.43+, `pennylane-lightning` 0.43+, PyTorch 2.0+, torchvision, Qiskit 2.3.1, `qiskit-machine-learning` 0.9.0, `qiskit-ibm-runtime` 0.45+.

Local execution (PennyLane main + ablations + noise):

```
python3 code/qtcl_v6_experiment.py          # main Penny
python3 code/qtcl_qiskit_experiment.py      # Qiskit bac
python3 code/qtcl_noise_experiment.py       # FakeBrisba
```

Hercules HPC (SLURM, all results reproduced here):

```
sbatch hercules/submit_quantics_main.slurm  # main + ab
sbatch hercules/submit_noise.slurm          # IBM FakeB
```

The Hercules jobs use 24–32 CPU cores and 48–64 GB RAM on the `standard` partition. Environment setup: `module load Miniconda3 && conda activate qtcl`.

IBM Quantum hardware (drop-in):

```
export IBM_QUANTUM=true
export IBM_QUANTUM_TOKEN=<token>
export IBM_BACKEND=ibm_brisbane
python3 code/qtcl_qiskit_experiment.py
```

Hyperparameters: all in the top-level constants of each script. Default configuration matches Table 2. Code, figures, and pre-computed results: https://github.com/danimap27/qtcl-paper

B Classical Baseline Details

The classical encoder architecture is identical to the QTCL encoder: $784 \xrightarrow{\text{Linear}} \xrightarrow{\text{ReLU}} \xrightarrow{\text{LN}} 32 \xrightarrow{\text{Linear}} \xrightarrow{\text{Tanh}} 4$. The classical head is $4 \xrightarrow{\text{Linear}} \xrightarrow{\text{ReLU}} 8 \xrightarrow{\text{Linear}} 2$. Total parameters: ≈25,000 (encoder: 24,932; head: 58). This ensures the comparison is fair: both models have the same input representation capacity.

C Quantum Circuit Parameters

PennyLane `StronglyEntanglingLayers`: each layer applies (R_Z, R_Y, R_Z) to each qubit (3 parameters per qubit per layer) plus entangling CNOT gates with a pattern that shifts by 1 qubit per layer (layer l uses range $[0, l \bmod n_q]$). Total: $(L_s + L_t) \times n_q \times 3 = 3 \times 4 \times 3 = 36$ parameters.

Qiskit `EfficientSU2`: applies (R_Y, R_Z) rotation blocks to each qubit per rep plus CX entanglement.

Parameters: $n_q \times 2 \times (\text{reps} + 1)$ per block. Shared ($L_s = 2$ reps): $4 \times 2 \times 3 = 24$; Task ($L_t = 1$ rep): $4 \times 2 \times 2 = 16$. Total: 40 parameters.

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